Skills Summary

Illegal Cancellation and Lost Solutions

Use this reference while completing your worksheet or when studying.

Skill 1 — Factor, never cancel

Rule: you may divide both sides by a quantity only if that quantity can never be zero. A trigonometric factor *can* be zero, so dividing by it silently deletes solutions.

Repair: move everything to one side, factor, and set each factor to zero.

$AB = A \implies A(B-1) = 0$, never $B = 1$ alone.

Example: Solve $2\cos^2\theta = \sqrt{3}\cos\theta$ for $0° \le \theta < 360°$.

Step 1. The same cosine appears on both sides, which is the tempting moment to divide — so collect and factor instead, keeping that cosine as a factor.

Step 2. $\cos\theta\,(2\cos\theta - \sqrt{3}) = 0$: $\cos\theta = 0$ or $\cos\theta = \dfrac{\sqrt{3}}{2}$. $\Rightarrow$ $\theta = \mathbf{30°, 90°, 270°, 330°}$.

Try it: Solve $2\sin^2\theta = \sqrt{2}\sin\theta$ for $0° \le \theta < 360°$.

check: $\theta = \mathbf{0°, 45°, 135°, 180°}$

Skill 2 — Keeping every solution

Three places solutions go missing:

- taking one square root: $u^2 = k$ means $u = \sqrt{k}$ *or* $u = -\sqrt{k}$;
- a multiple angle: for $k\theta$, multiply both interval endpoints by k before solving;
- declaring a factor impossible: $\cos\theta = 1$ and $\sin\theta = -1$ are attainable.

Example: Solve $2\cos 2\theta = \sqrt{3}$ for $0° \le \theta < 360°$.

Step 1. The angle inside is doubled, so the inside variable travels twice as far — widen its interval before solving, or half the answers never appear.

Step 2. $u = 2\theta$ on $[0°, 720°)$: $\cos u = \dfrac{\sqrt{3}}{2}$ at $u = 30°, 330°, 390°, 690°$; halve each. $\Rightarrow$ $\theta = \mathbf{15°, 165°, 195°, 345°}$.

Try it: Solve $2\sin 3\theta = \sqrt{3}$ for $0° \le \theta < 180°$.

check: $\theta = \mathbf{20°, 40°, 140°, 160°}$

Skill 3 — Testing for extraneous roots

Squaring is not reversible. If $x = y$ then $x^2 = y^2$, but not the other way round: -1 and 1 have the same square. So squaring both sides can *invent* roots.

Repair: squaring is allowed, but every candidate must then be substituted into the *original* equation and kept only if both sides agree.

Example: Solve $\cos\theta + 1 = \sin\theta$ for $0° \le \theta < 360°$.
Step 1. Two different ratios and no factor in common, so squaring is the way in — but it obliges me to test every candidate afterwards.
Step 2. Squaring gives $2\cos\theta(\cos\theta + 1) = 0$, so the candidates are 90°, 270° and 180°. Testing: 90° gives $1 = 1$, 180° gives $0 = 0$, but 270° gives $1 = -1$. $\Rightarrow$ $\theta = \mathbf{90°}, \mathbf{180°}$.

Try it: Solve $\sin\theta + \cos\theta = 1$ for $0° \le \theta < 360°$.

check: $\theta = \mathbf{0°}, \mathbf{90°}$